Autonomous and Intelligent Systems
Internship report

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1 Evaluating LoRa Comunication Protocol

2 Setup

In this section is presented how to setup correctly the OS for the RaspberryPi, the repository and the libraries used by the proposed solution.
After setting up the Raspberry Pi following the official guide it's necessary to enable SPI (Serial Peripheral Interface) selecting "Interface" and then "SPI" after using the following command.

```
sudo raspi-config
```

Then it's necessary to download the WiringPi library used to access GPIO with high performance. To do so requires the following command:

```
wget https://github.com/WiringPi/WiringPi/releases/download/3.16/wiringpi_3.16_armhf.deb
sudo dpkg -i wiringpi_3.16_armhf.deb
```

Lastly it's necessary to modify two files inside the LoRa-RaspberryPi folder:

- Makefile: #PYTHONVER=python3.11 needs to match system's Python version:
- lora.c: it's necessary to add #define PY_SSIZE_T_CLEAN on the first line for compatibility with new versions of Python.

To check out which version of Python it's installed use this command:

```
python3 --version
```

These two changes are already been made in the updated repository with the proposed solution inside the "src" folder.
Finally it's necessary to compile with the following command:

```
make all
```

3 Proposed solution

In this section is presented the proposed solution, starting with its folder structure then explaining the code for the sender and the receiver. Lastly the code for the analysis will be presented.

3.1 Folder structure

Inside the repository there are three folder:

- report: this folder can be ignored since it has been used to store the files needed for this report.
- src: inside this folder there's the code for the receiver and the sender, then the libraries used by both. There's also a utils folder with helper functions for logging and converting digits.
- analysis: inside this folder there's a notebook with the code for the analysis that will be explained later.

3.2 Sender

Insider the sender folder there are two files:
hf-sender.py: this file was used to test the maximum message sending capacity. It wasn't used during the last tests, the reasons will explained later.
During the tests with hf-sender it has been observed that the messages were corrupted or the receiver didn't receive them correctly.
After reading this article the code has been updated in a new file called sender.py In this new file the messages were delayed based on the airtime.
To calculate the airtime it has been used the formula defined in Semtech's LoRa Modem Designer's Guide (AN1200.13).
This website visually explain how airtime is influenced by factor like payload size and data rate.
To start sending messages use the following command inside the "src" folder:

```
python -m sender.sender
```

3.3 Receiver

Insider the receiver folder there are two files:
receiver.py: this file was used to test if the receiver could actually receive and log the message. After the first test it has been observed that looping through a fixed range of values isn't the best approach since the sender and the receiver can't synchronize.
After this consideration it has been decided to choose a fixed amount of time, in this case 2 minutes, to send and receive messages.
To start receiving messages use the following command inside the "src" folder:

```
python -m receiver.hf-receiver
```

3.4 Utils

Inside the utils folder there are two files:

- conversion.py: contains two helper function that convert a positive number to its digit representation in base b.
- loggin.py: contains one helper function that create a logger and setup the folders to save the data from the communicator.

The first file has been created to solve the following problem: since the LoRaRaspberryPi library only transmit bytes of data it's necessary to convert the counter to a list of digits in base 255.
It's has been used a list of digits to be able to send a counter greater than 255.

3.5 Analysis

In this section will be presented the data analysis based on the tests made outside the SMACT 3 laboratory.
In total 7 test has been done:

- first test: communication in the parking lot of the laboratory, 100 meters. LoRa spreading factor set to 7.
- second test: same as the first but with a spreading factor set to 12.
- third test: from the water storage outside the canteen to the trees, 100 meters LoRa spreading factor set to 12.
- fourth test: same as the first but between the RaspberryPis there were more trees.

- fifth test: from the tree next to the canteen to "Canale Ledra". LoRa spreading factor set to 12.
- sixth test: communication from Autonomous System laboratory to Cybersecurity laboratory. LoRa spreading factor set to 10
- seventh test: communication inside the Autonomous System laboratory. LoRa spreading factor set to 10.

The analysis starts with the installation of the required packages and the definition of the function already presented before.
After that the data has been cleaned revealing 5 error in the sender logger and 1 invalid message in the receiver logger. Since it's a small amount these errors can be ignored.
The first part of the analysis focuses on verifying how many messages has been successfully received. The pie chart of the message outcomes per test appears in figure number 1.

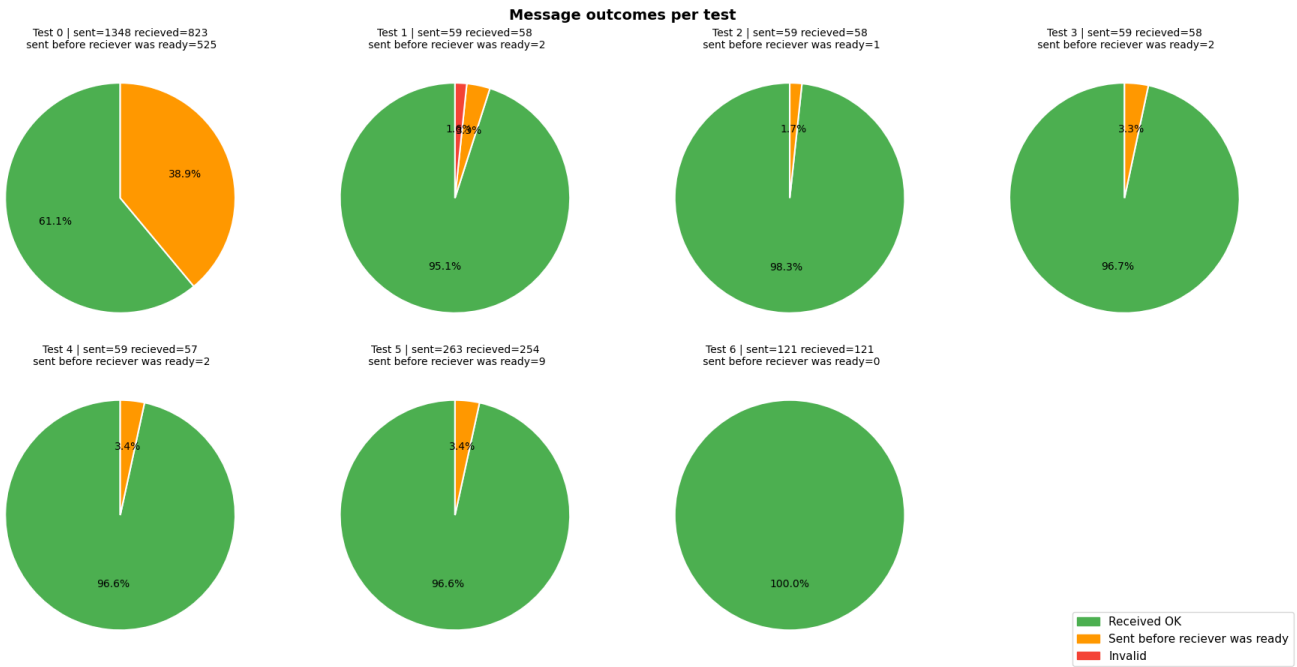

Figure 1: Pie charts of the message outcomes per test

Since the tests were synchronized manually it's necessary to consider that sometimes the sender and the receiver weren't synchronized. The messages that were sent before the receiver was ready won't be considered as errors but will be considered as a category per se instead.
Given this assumption the plot highlight that only one message in the second test was invalid. It is necessary to remember that there were 5 errors in the sender's log. In particulary, there are one error in second, third and fourth test and two errors in the fifth one.
The analysis continues with checking on latencies. In particular the focus in on the median, maximum and 99-th percentile to highlight possible spikes and to define acceptable bounds. For this part will be presented the latency per message in figure 2, the latency distribution per test in figure 3 and the latency heatmap in figure 4.

askjd

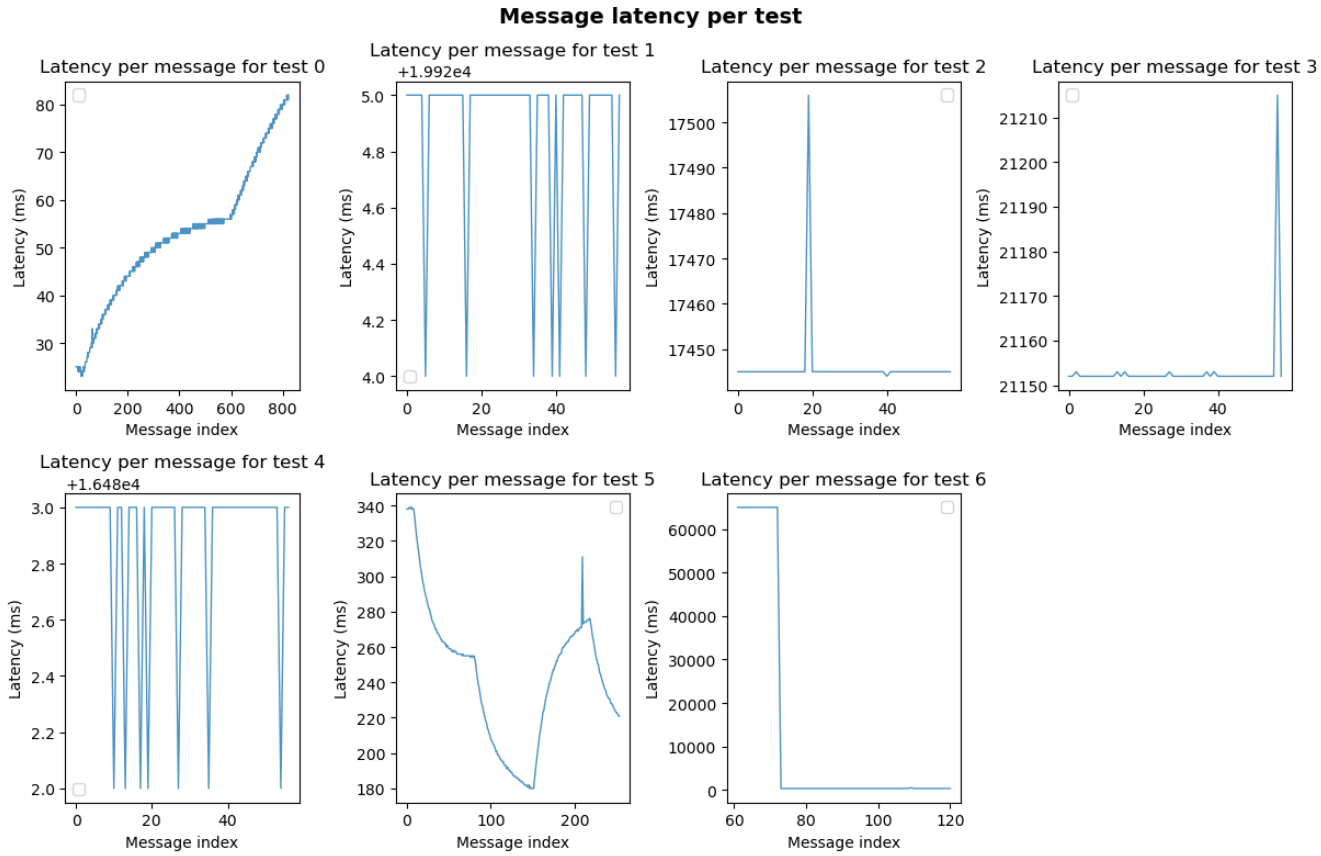

Figure 2:

askjd
askjd

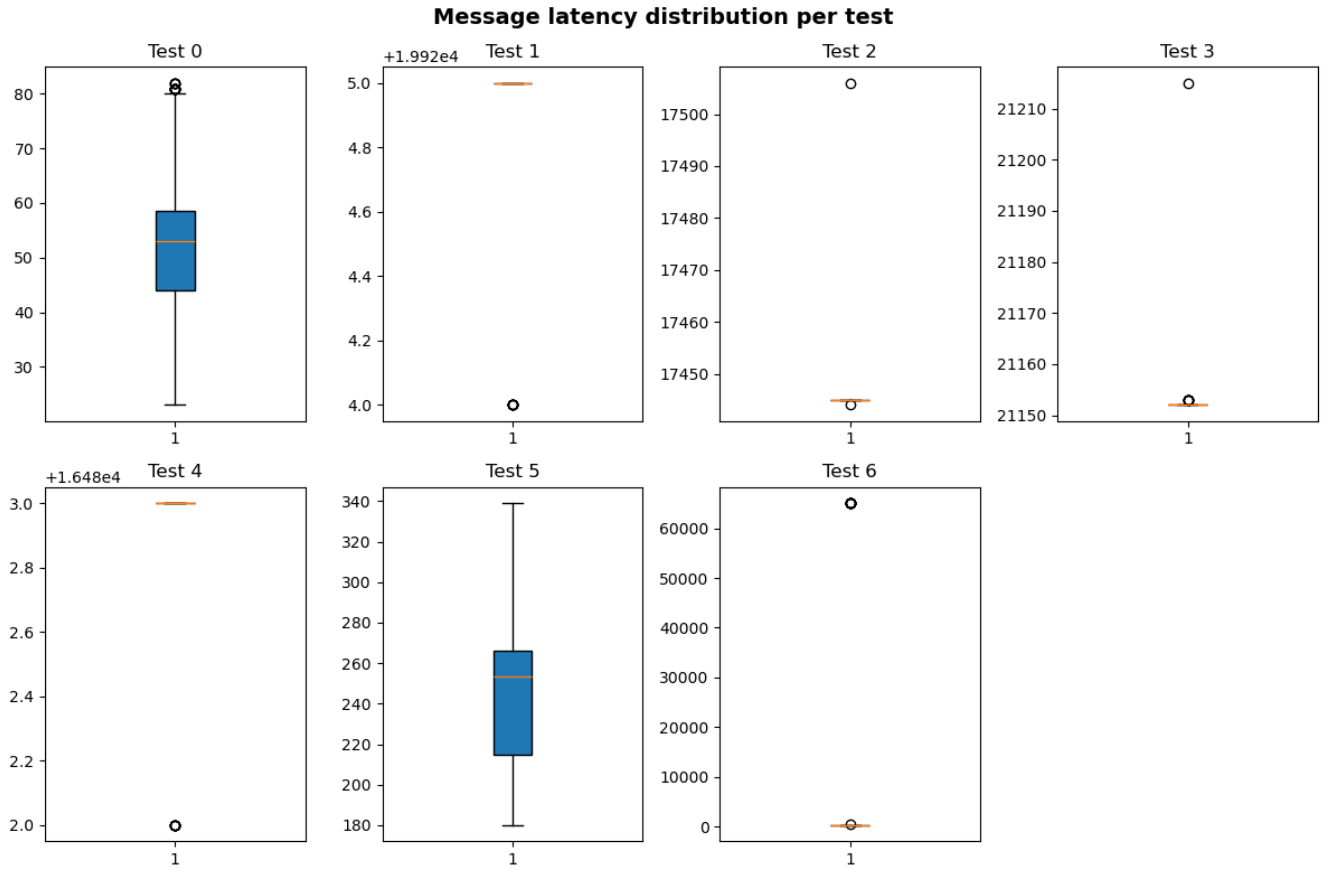

Figure 3:

4 Conclusion

Conclusion

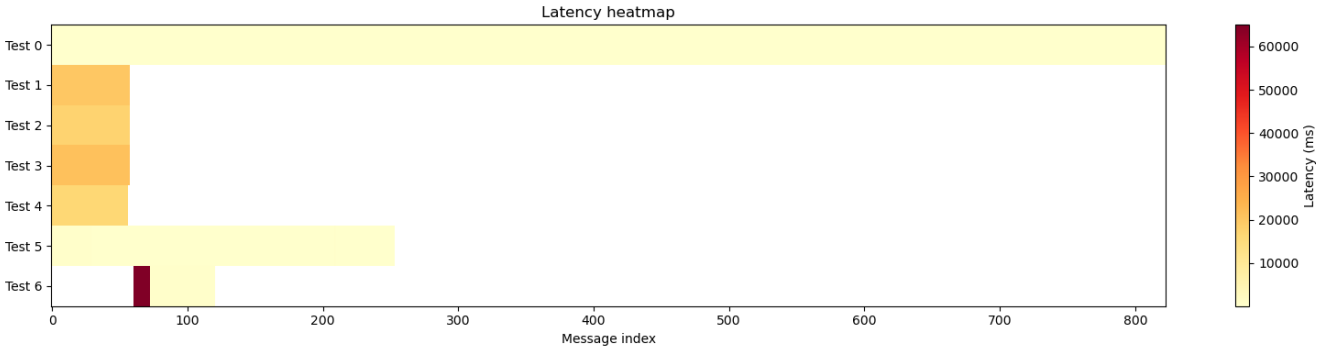

Figure 4: Pie charts of the message outcomes per test